

HAT-P-3b

R-1446

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-3_160704 · created 2026-08-01T09:05:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457573.7541 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.012 +/- 0.042
Transit depth (Rp/Rs)^2	0.015 +/- 0.1 [%]
Orbital Inclination (inc)	88.49 +/- 2.22
Airmass coefficient 1 (a1)	1.73 +/- 0.36
Airmass coefficient 2 (a2)	-0.294 +/- 0.038
Scatter in the residuals of the lightcurve fit is	19.75 %
Best Comparison Star	None
Optimal Aperture	32.96
Optimal Annulus	60.05
Transit Duration (day)	0.09 +/- 0.11

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.012 $\pm$ 0.042 vs 0.1063 (deviation 1.04 σ)
Duration fit vs published	0.09 d vs 0.0867 d (deviation 0.01 σ)
Mid-time O-C	-3.4 min
Depth SNR	0.1
Residual scatter	19.75 %

Light curve

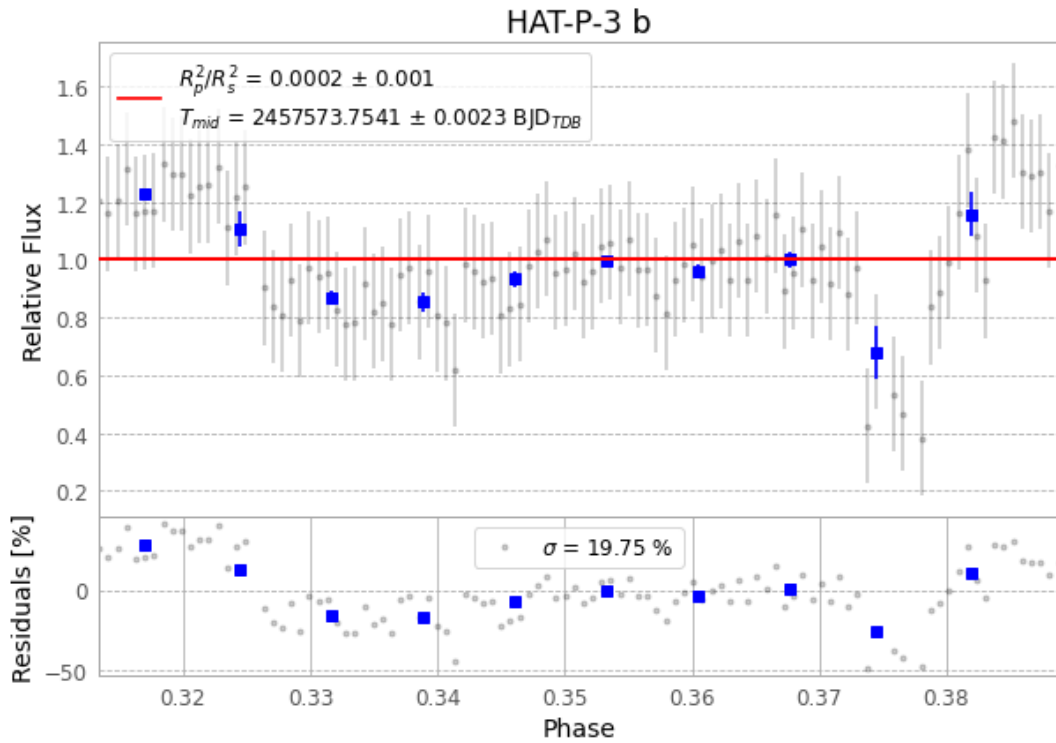


Figure 1 — FinalLightCurve_HAT-P-3 b_2016-07-04.png (SHA-256 10357d42a63df26f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [237, 322], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 33c0cb0d28f53291...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

106 FITS frames (70 MB), HATP-3160705035412.FITS → HATP-3160705090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-160704.log (b0d4035107bc...), FinalLightCurve_HAT-P-3 b_2016-07-04.png (10357d42a63d...), FinalLightCurve_HAT-P-3 b_2016-07-04.csv (5b2fcd4814a5...)

Compute resources

Wall time 420.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 09:05Z.